We can also determine the centre of gravity of a uniform lamina of even an irregular shape by pivoting it at various points by trial and error. The lamina remains horizontal when pivoted at the point where the net gravitational force acts, which is the centre of gravity as shown in Figure 5.17. When a body is supported at the centre of gravity, the sum of the torques acting on all the point masses of the rigid body becomes zero. Moreover the weight is compensated by the normal reaction force exerted by the pivot. The body is in static equilibrium and hence it remains horizontal.

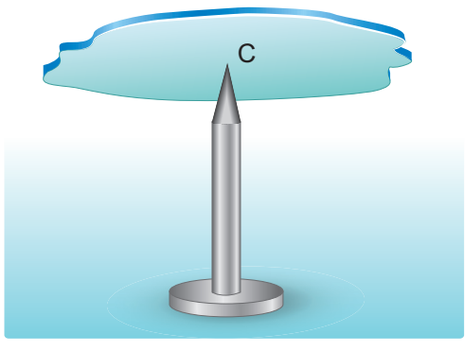


Figure 5.17. Determination of centre of gravity of plane lamina by pivoting

There is also another way to determine the centre of gravity of an irregular lamina. If we suspend the lamina from different points like P, Q, R as shown in Figure 5.18, the vertical lines PP', QQ', RR' all pass through the centre of gravity. Here, reaction force acting at the point of suspension and the gravitational force acting at the centre of gravity cancel each other and the torques caused by them also cancel each other.

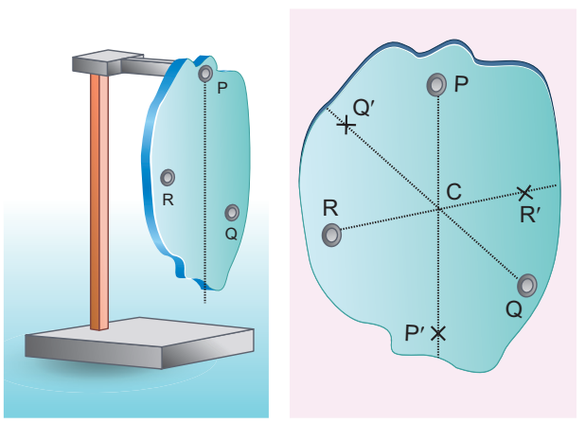


Figure 5.18 Determination of centre of gravity of plane lamina by suspending

5.3.5 Bending of Cyclist in Curves

Let us consider a cyclist negotiating a circular level road (not banked) of radius r with a speed v. The cycle and the cyclist are considered as one system with mass m. The centre gravity of the system is C and it goes in a circle of radius r with centre at O. Let us choose the line OC as X-axis and the vertical line through O as Z-axis as shown in Figure 5.19.

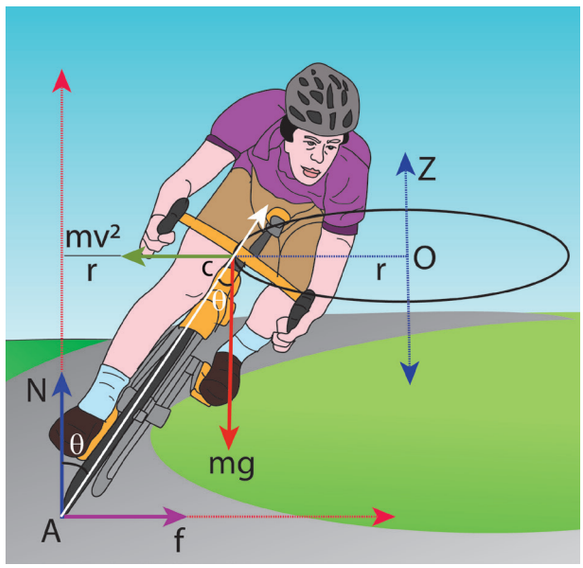


Figure 5.19 Bending of cyclist

UNIT-5(XI-Physics_Vol-1).indd 232 06-01-2022 17:46:59